AMERY ICE SHELF BOREHOLE MICROCAT AND CTD DATA, 2006/2007 SEASON - DATA PROCESSING AND QUALITY

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INTRODUCTION

This report summarises the latest AMISOR microcat data, season 2006/2007, collected from oceanographic equipment deployed through the Amery Ice Shelf boreholes, including details of data processing and data quality. Temperature/salinity/pressure time series data were downloaded from four strings of Seabird SBE37IM microcats at AMISOR boreholes AM01 (2 years of data) and AM02, AM03 and AM04 (a ~year of data each).

Table 1: Station details for AMISOR season 2006/2007 borehole microcats. Note that borehole absolute locations change in time, as the ice shelf is in motion. For the microcats, positive clock errors indicate microcat clocks were fast. Instrument pressures/depths are mean values over the data dates.

borehole	position at given date		microcat	mean instrument pos.		data dates		rec.	clock
	latitude	longitude		depth (m)	pressure (dbar)	start	end	interval (min.)	error (sec.)
AM01	69°24.990'S	71°29.284'E	1969	436.2	441.1	31-Dec-2004	06-Dec-2006	30	+313
AM01	(on 05-Dec-2006)		1970	574.8	581.5	31-Dec-2004	06-Dec-2006	30	+304
AM01	"	"	1971	734.2	743.0	31-Dec-2004	06-Dec-2006	30	+351
AM02	69°41.444'S	72°42.464'E	1623	332.8	336.5	22-Feb-2006	15-Dec-2006	30	+86
AM02	(on 15-Dec-2006)		1624	554.5	560.9	22-Feb-2006	16-Dec-2006	30	+206
AM02	"	"	1174	761.3	770.5	22-Feb-2006	16-Dec-2006	30	+85
AM03	xx°xx.xx'S	xx°xx.xx'E	4055	653.6	661.4	27-Dec-2005	03-Dec-2006	30	+162
AM03	(on xx-xxx-xxxx)		4054	861.5	872.2	27-Dec-2005	03-Dec-2006	30	+127
AM03	"	"	3883	1211.1	1227.1	27-Dec-2005	03-Dec-2006	30	+84
AM04	69°53.762'S	70°17.819'E	1972	543.5	549.8	11-Jan-2006	25-Nov-2006	30	+141
AM04	(on 25-Nov-2006)		1973	684.3	692.5	11-Jan-2006	25-Nov-2006	30	+177
AM04	"	"	1974	803.7	813.5	11-Jan-2006	25-Nov-2006	30	+91

MICROCAT DATA PROCESSING

The microcat moorings have already been described in previous reports. The mooring at AM01 was deployed in January 2002; January 2001 for AM02; December 2005 for AM03; and January 2006 for AM04. All latest microcat data downloaded during the 2006/2007 season (Table 1) were assigned the same decimal time scheme consistent with previous data i.e. decimal days as counted from midnight on December 31st 2000 (so, e.g. midday on January 1st 2001 = 0.5 decimal time; midday on January 1st 2002 = 365.5).

The microcats were downloaded by Dave Rasch using the SeaBird terminal program Seaterm, and instrument clock errors were noted at the time (Table 1). Clocks were reset to GPS time for all sets of microcats, following data download and prior to reinitialisation of logging.

Manufacturer supplied calibrations (May/June 2001 for AM01 instruments, July/August/September 2000 for AM02 instruments, June 2005 for AM03 instruments [except October 2004 for microcat-3883 pressure], and February 2003 for AM04 instruments) were applied internally by the microcats, and calibrated data were output. The data were then processed, as follows:

* The files were padded at the start and end, data gaps were checked for and filled, and decimal times were calculated.

* For AM01, pressure data were bad for all three instruments. A constant pressure value was selected for each AM01 instrument (Table 1), from inspection of data near the start of the deployment in January 2002. These pressure values were used to recalculate conductivity, and thence in the salinity calculations for each instrument.

* For AM02, AM03 and AM04, a mean pressure value was calculated for the year's data for each microcat (Table 1).

* Decimal times were “compressed” linearly throughout the time series to correct for clock error (Table 1).

MICROCAT DATA QUALITY NOTES

The following bad data were removed from the files:

* AM02, microcat 1623 - conductivity/salinity data spike removed for data points 10687-10688
  - conductivity/salinity data spike removed for data points 11831-11840

* AM02, microcat 1174 - several bad data scans occur from 8th to 9th May 2006, due to either logging or download errors; null values inserted for missing data
  - conductivity data still bad, as per previous seasons (sensor went bad in March 2004); all conductivity and salinity data removed

* AM03, microcat 4055 - conductivity/salinity data spike removed for data point 9409

* AM03, microcat 3883 - conductivity/salinity data spike removed for data point 10119

* AM04, microcat 1972 - bad conductivity/salinity data removed for data points 12094-12201
  - bad conductivity/salinity data removed for data points 12463-12500
  - bad conductivity/salinity data removed for data points 14104-14110

* AM04, microcat 1974 - conductivity/salinity data spike removed for data points 1406-1408
  - conductivity/salinity data spike removed for data point 4938
  - conductivity/salinity data spike removed for data points 5208-5210
  - conductivity/salinity data spike removed for data point 14855

The following data appear suspect (but not removed from the files):

* AM01, microcat 1969 - for first ~1.5 days, conductivity (and salinity) values slowly drift up, by ~0.03 mS/cm (Figure 1)

* AM04, microcat 1974 - from start to decimal time ~1864.23 (i.e. first ~23 days), pressure slowly drops by ~1.2 dbar (Figure 5) (any error to conductivity/salinity is not significant)

When resetting microcat 1174 at AM02, Dave Rasch noted that the instrument gave a low battery warning. This microcat appears to be on its last legs.

Microcat data are plotted in Figures 1 to 5. For the deepest microcat at AM03 (microcat 3883) (Figure 3), an abrupt transition in water properties occurs at decimal day ~2100. Given the similar nature of the data from the previous summer (i.e. at the start of the time series in Figure 3), there is no reason to suspect this data feature.

From conductivity and temperature sensor calibration drift estimates in a previous data report for 2003/2004 data (i.e. 0.0003 (PSS78) **per month** for salinity, and 0.0005^{o}C **per year** for temperature), we might expect the following calibration drifts at the time of the latest 2006/2007 data downloads:

AM01 (in water for 5 years)	total S drift = 0.018 (PSS78)	total T drift = 0.0025^{o}C
AM02 (in water for 6 years)	total S drift = 0.022 (PSS78)	total T drift = 0.0030^{o}C
AM03 (in water for 1 year)	total S drift = 0.004 (PSS78)	total T drift = 0.0005^{o}C

For AM04, manufacturer sensor calibrations were from February 2003, and the following total drifts might be expected at the time of the latest 2006/2007 data downloads (despite only 1 year in the water):

AM04 (after 4 years)	total S drift = 0.014 (PSS78)	total T drift = 0.0020^{o}C

Pressure drift for the AM02, AM03 and AM04 microcats (Figure 5) is of the order 1 dbar or less over the latest year of data, typically decreasing with time. This is of the same order as the expected pressure sensor drift as quoted by the manufacturer, however from the microcat data alone it is difficult to separate the contributing factors to this drift i.e. calibration drift, and/or physical change in instrument position due to ice shelf change.

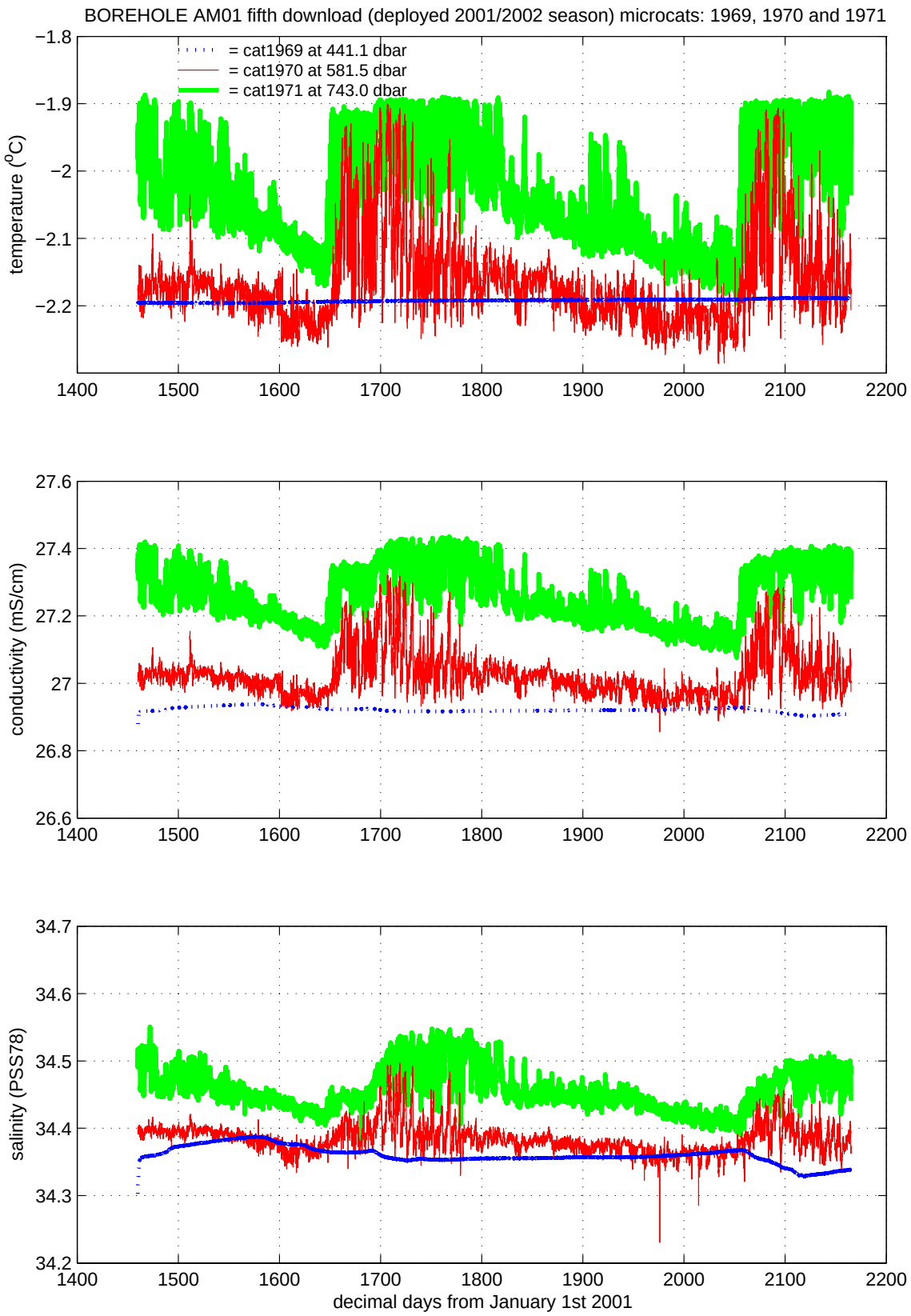


Figure 1: Borehole AM01 microcat data downloaded in 2006/2007 season.

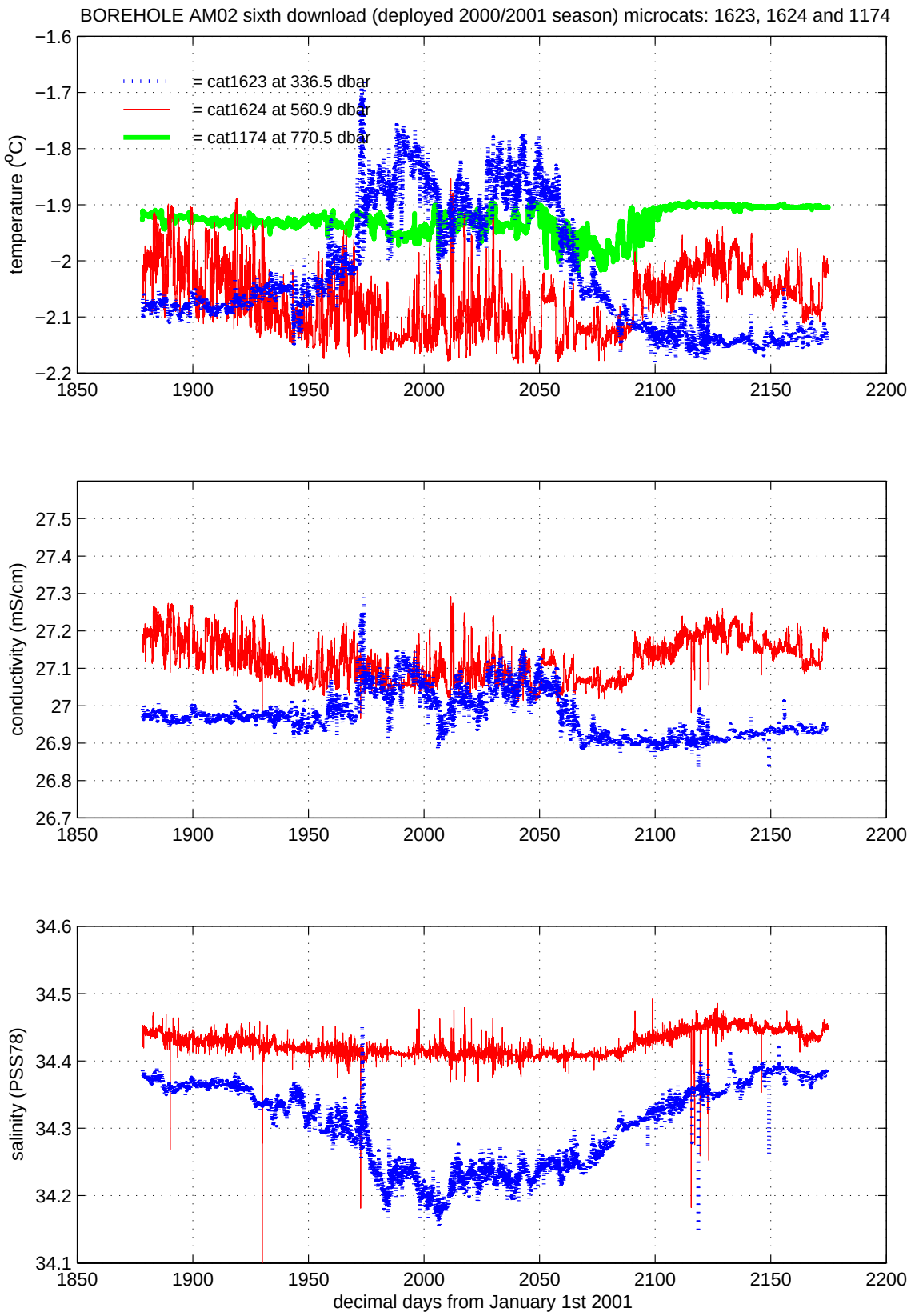


Figure 2: **Borehole AM02 microcat data downloaded in 2006/2007 season.**

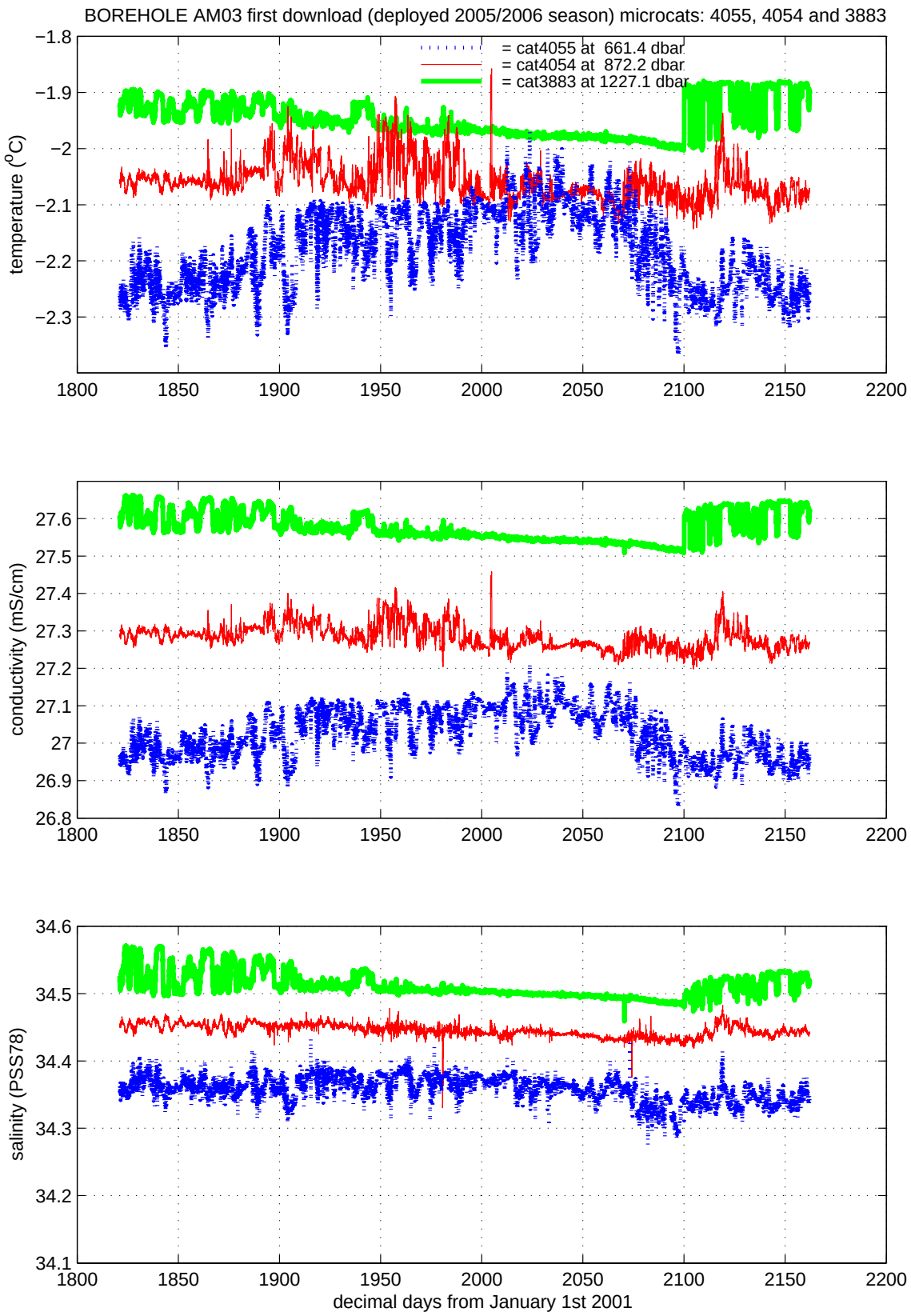


Figure 3: **Borehole AM03 microcat data downloaded in 2006/2007 season.**

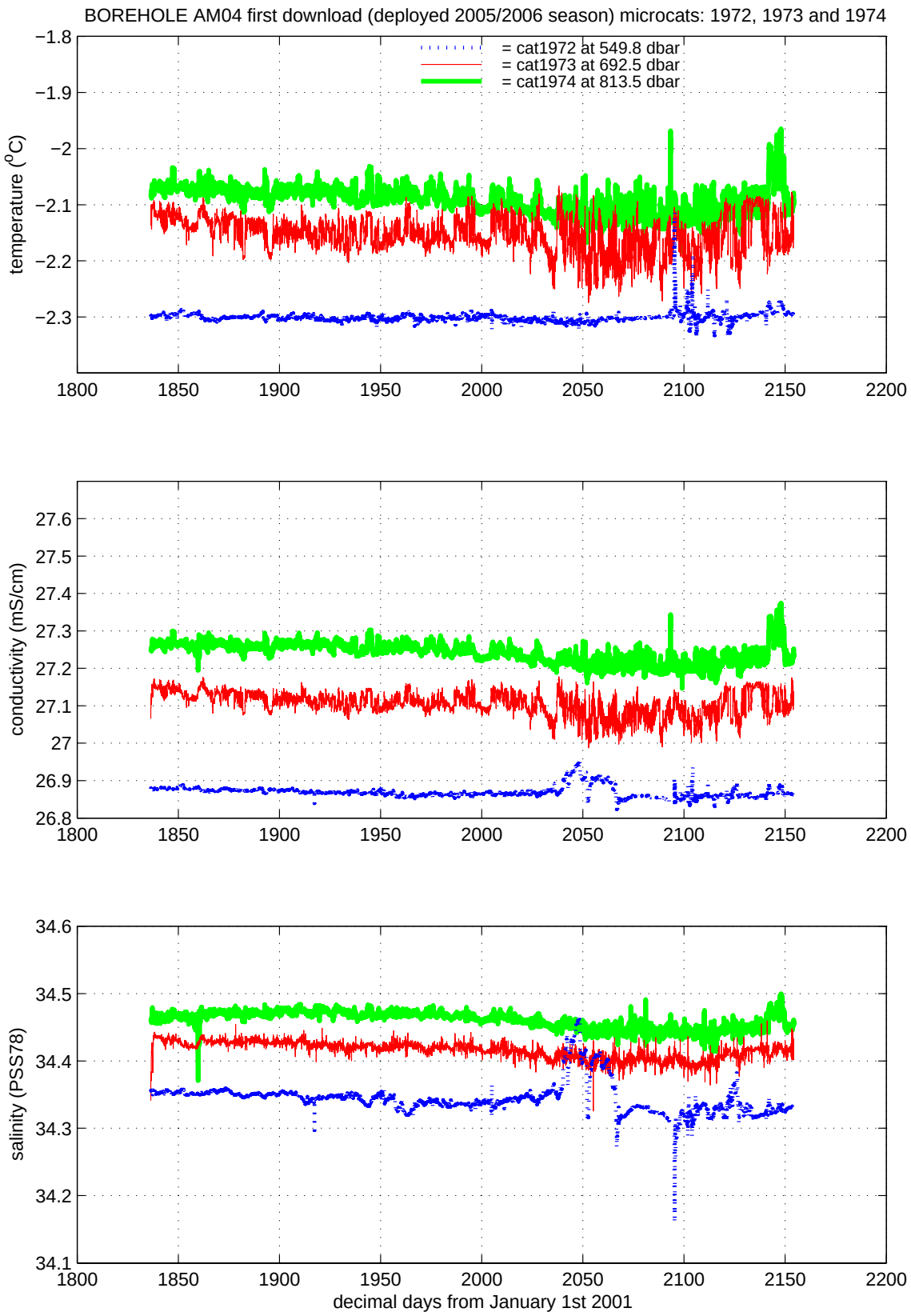
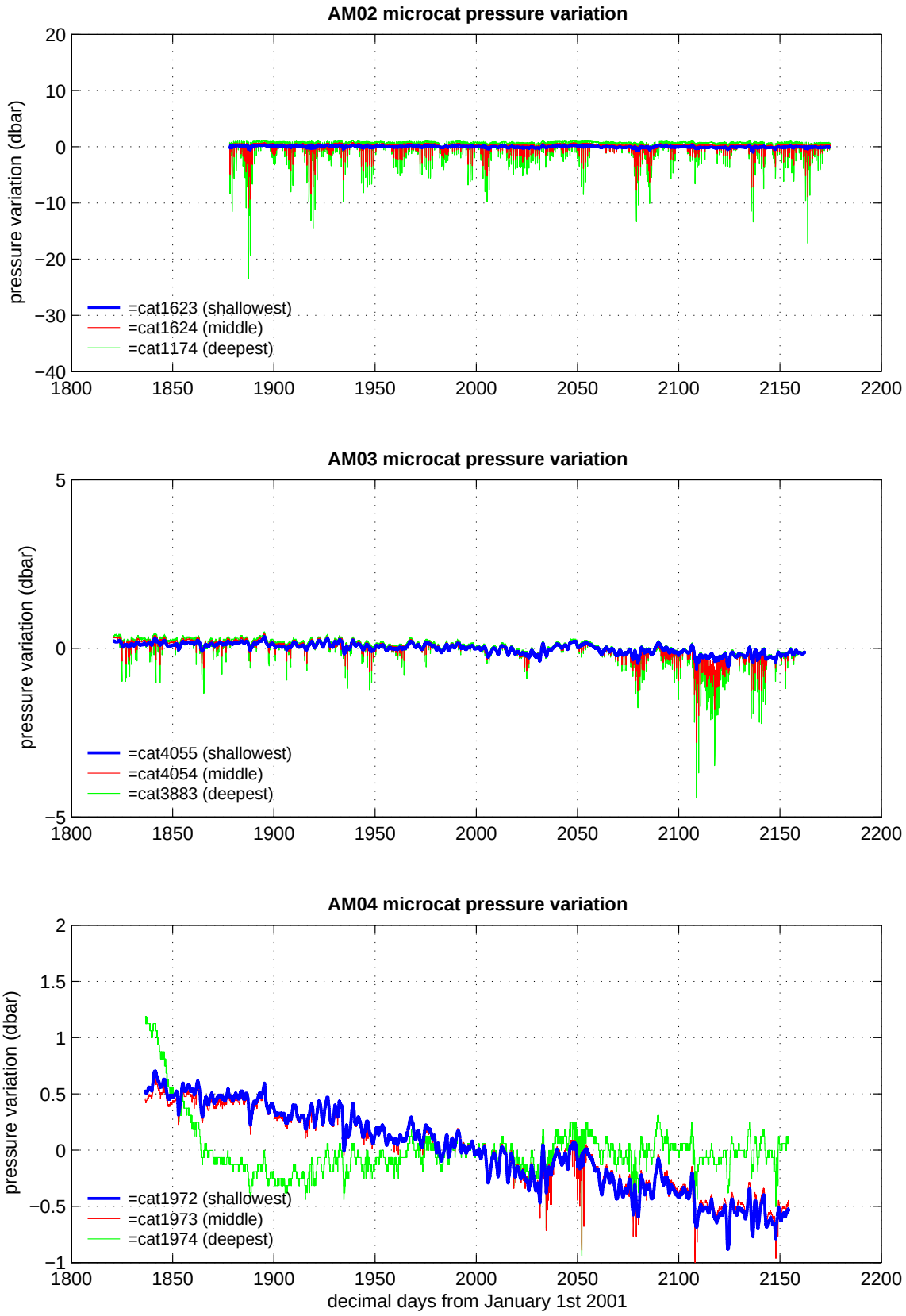


Figure 4: Borehole AM04 microcat data downloaded in 2006/2007 season.



<u>Figure 5:</u> Pressure sensor data from boreholes AM02, AM03 and AM04 microcat data downloaded in 2006/2007 season. Plotted pressures are variations about the mean values.

DATA FILES

1. Microcat data from latest downloads

New microcat data from boreholes AM01, AM02, AM03 and AM04 are contained in text and matlab format files, as follows:

AM01
Data from the fifth data download (i.e. 31/12/2004-06/12/2006) are contained in the files:
cat1969_5.dat, cat1970_5.dat, cat1971_5.dat (text files), and
cat1969_5.mat, cat1970_5.mat, cat1971_5.mat (matlab files).
READMEcatam01_5 details these microcat file formats.

AM02
Data from the sixth data download (i.e. 22/02/2006-16/12/2006) are contained in the files:
cat1623_6.dat, cat1624_6.dat, cat1174_6.dat (text files), and
cat1623_6.mat, cat1624_6.mat, cat1174_6.mat (matlab files).
READMEcatam02_6 details these microcat file formats.

AM03
Data from the first data download (i.e. 27/12/2005-03/12/2006) are contained in the files:
cat4055_1.dat, cat4054_1.dat, cat3883_1.dat (text files), and
cat4055_1.mat, cat4054_1.mat, cat3883_1.mat (matlab files).
READMEcatam03_1 details the microcat file formats.

AM04
Data from the first year data download (i.e. 11/01/2006-25/11/2006, including the initial 5 day download fromthe 2005/2006 season) are contained in the files:
cat1972_1.dat, cat1973_1.dat, cat1974_1.dat (text files), and
cat1972_1.mat, cat1973_1.mat, cat1974_1.mat (matlab files).
READMEcatam04_1 details the microcat file formats.

2. Unified microcat data files

The updated unified files (i.e. all data from initial deployment time, updated with data from the latest download) are contained in the following files:

AM01 (i.e. 15/01/2002-06/12/2006)
cat1969_all.dat, cat1970_all.dat, cat1971_all.dat (text files), and
cat1969_all.mat, cat1970_all.mat, cat1971_all.mat (matlab files).
READMEcatam01_all details these microcat file formats.
catamisor1all.eps is a time series plot.

AM02 (i.e. 06/01/2001-16/12/2006)
cat1623_all.dat, cat1624_all.dat, cat1174_all.dat (text files), and
cat1623_all.mat, cat1624_all.mat, cat1174_all.mat (matlab files).
READMEcatam02_all details these microcat file formats.
catamisor2all.eps and catamisor2allp.eps are time series plots.

AM03 (i.e. 27/12/2005-03/12/2006)
cat4055_all.dat, cat4054_all.dat, cat3883_all.dat (text files), and
cat4055_all.mat, cat4054_all.mat, cat3883_all.mat (matlab files).
READMEcatam03_all details these microcat file formats.
catamisor3all.eps and catamisor3allp.eps are time series plots.

AM04 (i.e. 11/01/2006-25/11/2006)
cat1972_all.dat, cat1973_all.dat, cat1974_all.dat (text files), and

cat1972_all.mat, cat1973_all.mat, cat1974_all.mat (matlab files).
READMEcatam04_all details these microcat file formats.
catamisor4all.eps and catamisor4allp.eps are time series plots.

3. Station and summary info files

The files station.dat and borehole_data_summary.dat are updated to include the latest season's data.